

AERIS Expert Review Packet

Anomaly

Source / metric	openaq / PM10 (pm10)
Observed / expected baseline	93 / 49.7963 ug/m3
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-07-23 01:00 UTC
Location	29.6700, -95.1285
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	7 / 746	29.01 to 100; mean 70.9953	83.93 at 2026-07-23 00:55Z; 5 min before event
asos	Air temperature (temperature)	degC	7 / 746	24.7778 to 39; mean 30.0681	29 at 2026-07-23 00:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	7 / 906	0 to 360; mean 171.369	180 at 2026-07-23 00:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	7 / 928	0 to 8.74555; mean 3.0013	2.05778 at 2026-07-23 00:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	8 / 96	5919.35 to 5959.62; mean 5936.39	5931.74 at 2026-07-23 00:00Z; 1 h before event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	8 / 96	17.7446 to 2017.14; mean 666.612	980.668 at 2026-07-23 00:00Z; 1 h before event
noaa_gfs	Precipitable water (precipitable_water)	mm	8 / 96	32.3551 to 67.4723; mean 50.9081	58.4769 at 2026-07-23 00:00Z; 1 h before event
noaa_gfs	Surface pressure (surface_pressure)	hPa	8 / 96	1003.85 to 1014.98; mean 1008.94	1005.31 at 2026-07-23 00:00Z; 1 h before event
noaa_gfs	850-hPa temperature (t_850)	degC	8 / 96	17.875 to 24.7709; mean 20.9541	24.3609 at 2026-07-23 00:00Z; 1 h before event
noaa_gfs	10-m eastward wind (u_10m)	m/s	8 / 96	-4.5306 to 4.0994; mean 0.402	-1.3906 at 2026-07-23 00:00Z; 1 h before event
noaa_gfs	10-m northward wind (v_10m)	m/s	8 / 96	-6.0398 to 5.78799; mean -0.2277	-0.422297 at 2026-07-23 00:00Z; 1 h before event
openaq	Ozone (ozone)	ppm	17 / 1130	-0.003 to 0.092; mean 0.0281	0.044 at 2026-07-23 01:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 182	3 to 98; mean 36.9615	93 at 2026-07-23 01:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	75 / 2923	0 to 34.1; mean 9.783	20.4 at 2026-07-23 01:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 360	0 to 100; mean 44.7278	85 at 2026-07-23 01:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	5 / 360	40 to 98; mean 72.9694	86 at 2026-07-23 01:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	5 / 360	0 to 31.24; mean 0.361	11.53 at 2026-07-23 01:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	5 / 360	1007 to 1016; mean 1010.77	1007 at 2026-07-23 01:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	5 / 360	24.91 to 40.07; mean 30.7151	29.04 at 2026-07-23 01:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	5 / 360	0 to 360; mean 178.189	203 at 2026-07-23 01:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	5 / 360	0.4 to 7.61; mean 2.3504	4.73 at 2026-07-23 01:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	62 / 4454	0 to 149.3; mean 11.7877	11.5 at 2026-07-23 01:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 19:14Z; 5 h 45 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 19:14Z; 5 h 45 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	6 / 6	0.0233633 to 0.0312825; mean 0.0271	0.0311848 at 2026-07-22 19:14Z; 5 h 45 min before event
sentinel5p	CO granules available (s5p_co_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 19:14Z; 5 h 45 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	5 / 5	0.000163343 to 0.000396712; mean 0.0002	0.000396712 at 2026-07-22 19:14Z; 5 h 45 min before event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-07-22 19:14Z; 5 h 45 min before event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	2 / 2	5.0358e-05 to 5.87309e-05; mean 0.0001	5.0358e-05 at 2026-07-22 19:14Z; 5 h 45 min before event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	4 / 4	1 to 1; mean 1	1 at 2026-07-22 19:14Z; 5 h 45 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 19:14Z; 5 h 45 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	5 / 5	-2.41161e-06 to 9.61895e-05; mean 0	-2.41161e-06 at 2026-07-22 19:14Z; 5 h 45 min before event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-07-22 19:14Z; 5 h 45 min before event
tceq	Carbon monoxide (co)	ppm	3 / 195	0 to 0.8; mean 0.1574	0.1 at 2026-07-23 01:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	12 / 716	-2.6 to 44.3; mean 6.4525	8.3 at 2026-07-23 01:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 191	-1 to 1.6; mean -0.2115	0.2 at 2026-07-23 01:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	07-21 12Z 68.54, 18Z 39.33, 07-22 00Z 61.22, 06Z 78.48, 12Z 59.36, 18Z 42.31, 07-23 00Z 69.74, 06Z 75.93, 12Z 74.27, 18Z 86.6, 07-24 00Z 93.19, 06Z 90.42, 12Z 91.86
asos	Air temperature (temperature)	07-21 12Z 29.93, 18Z 35.47, 07-22 00Z 30.65, 06Z 27.1, 12Z 31.74, 18Z 37.83, 07-23 00Z 31.46, 06Z 29.11, 12Z 29.24, 18Z 28.07, 07-24 00Z 25.96, 06Z 26, 12Z 27.21
asos	Wind direction (wind_direction)	07-21 12Z 271.5, 18Z 229.4, 07-22 00Z 186.8, 06Z 239.9, 12Z 289.3, 18Z 213.3, 07-23 00Z 141.1, 06Z 58.29, 12Z 53.85, 18Z 162.2, 07-24 00Z 152.2, 06Z 100.3, 12Z 115.4
asos	Wind speed (wind_speed)	07-21 12Z 3.783, 18Z 2.414, 07-22 00Z 2.254, 06Z 2.931, 12Z 3.799, 18Z 2.691, 07-23 00Z 2.144, 06Z 2.986, 12Z 4.676, 18Z 2.889, 07-24 00Z 3.377, 06Z 2.144, 12Z 3.087
noaa_gfs	500-hPa geopotential height (gh_500)	07-21 18Z 5932, 07-22 00Z 5929, 06Z 5934, 12Z 5927, 18Z 5939, 07-23 00Z 5929, 06Z 5942, 12Z 5926, 18Z 5934, 07-24 00Z 5940, 06Z 5957, 12Z 5946
noaa_gfs	Planetary boundary layer height (pbl_height)	07-21 18Z 1232, 07-22 00Z 775.9, 06Z 115.1, 12Z 230.8, 18Z 1741, 07-23 00Z 533.6, 06Z 396, 12Z 367.9, 18Z 1228, 07-24 00Z 530.7, 06Z 535.8, 12Z 313.1
noaa_gfs	Precipitable water (precipitable_water)	07-21 18Z 35.92, 07-22 00Z 40.6, 06Z 45.58, 12Z 52.41, 18Z 55.65, 07-23 00Z 60.49, 06Z 46.08, 12Z 50.49, 18Z 62.67, 07-24 00Z 64.11, 06Z 49.81, 12Z 47.08
noaa_gfs	Surface pressure (surface_pressure)	07-21 18Z 1011, 07-22 00Z 1008, 06Z 1008, 12Z 1009, 18Z 1008, 07-23 00Z 1005, 06Z 1008, 12Z 1008, 18Z 1008, 07-24 00Z 1008, 06Z 1013, 12Z 1013
noaa_gfs	850-hPa temperature (t_850)	07-21 18Z 22.23, 07-22 00Z 21.82, 06Z 22.45, 12Z 21.72, 18Z 21.78, 07-23 00Z 23.96, 06Z 21.64, 12Z 20.2, 18Z 19.26, 07-24 00Z 19.9, 06Z 18.05, 12Z 18.43
noaa_gfs	10-m eastward wind (u_10m)	07-21 18Z 2.326, 07-22 00Z 0.9697, 06Z 2.346, 12Z 3.066, 18Z 0.5057, 07-23 00Z -0.2256, 06Z -1.323, 12Z -0.4592, 18Z -1.793, 07-24 00Z 0.9149, 06Z -0.9048, 12Z -0.5983
noaa_gfs	10-m northward wind (v_10m)	07-21 18Z -0.7232, 07-22 00Z 0.000791, 06Z 1.277, 12Z -0.5226, 18Z -2.413, 07-23 00Z 0.694, 06Z -3.074, 12Z -3.549, 18Z -5.132, 07-24 00Z 4.71, 06Z 3.419, 12Z 2.581
openaq	Ozone (ozone)	07-21 12Z 0.01323, 18Z 0.0444, 07-22 00Z 0.02872, 06Z 0.006479, 12Z 0.01595, 18Z 0.06592, 07-23 00Z 0.03939, 06Z 0.02738, 12Z 0.02727, 18Z 0.02748, 07-24 00Z 0.02119, 06Z 0.0122, 12Z 0.005941
openaq	PM10 (pm10)	07-21 12Z 41.7, 18Z 49.09, 07-22 00Z 46.83, 06Z 42.2, 12Z 45.33, 18Z 51.67, 07-23 00Z 55, 06Z 30.56, 12Z 25.22, 18Z 16.39, 07-24 00Z 19.61, 06Z 36.94, 12Z 40
openaq	PM2.5 (pm25)	07-21 12Z 7.019, 18Z 5.717, 07-22 00Z 6.124, 06Z 6.039, 12Z 6.288, 18Z 9.13, 07-23 00Z 10.72, 06Z 18.24, 12Z 17.79, 18Z 8.624, 07-24 00Z 7.879, 06Z 11.47, 12Z 13.39
openweather	Cloud cover (cloud_cover)	07-21 12Z 1.375, 18Z 1.806, 07-22 00Z 3.267, 06Z 21.53, 12Z 11.6, 18Z 56.93, 07-23 00Z 76.31, 06Z 66.06, 12Z 96.9, 18Z 96.73, 07-24 00Z 93.47, 06Z 10.77, 12Z 3.2
openweather	Relative humidity (humidity)	07-21 12Z 76.71, 18Z 48, 07-22 00Z 61.2, 06Z 79.5, 12Z 70.5, 18Z 49.03, 07-23 00Z 70.03, 06Z 76.65, 12Z 77.23, 18Z 77, 07-24 00Z 93.63, 06Z 93.8, 12Z 95.2
openweather	Precipitation rate (precipitation)	07-21 12Z 0, 18Z 0, 07-22 00Z 0, 06Z 0, 12Z 0, 18Z 0.006, 07-23 00Z 1.453, 06Z 0, 12Z 0, 18Z 0.3917, 07-24 00Z 2.407, 06Z 0.1017, 12Z 0.122
openweather	Surface pressure (pressure)	07-21 12Z 1013, 18Z 1011, 07-22 00Z 1010, 06Z 1010, 12Z 1011, 18Z 1008, 07-23 00Z 1009, 06Z 1010, 12Z 1010, 18Z 1009, 07-24 00Z 1012, 06Z 1015, 12Z 1016
openweather	Air temperature (temperature)	07-21 12Z 29.5, 18Z 36.46, 07-22 00Z 31.98, 06Z 27.45, 12Z 31.06, 18Z 38.97, 07-23 00Z 32.77, 06Z 29.84, 12Z 29.6, 18Z 30.21, 07-24 00Z 25.83, 06Z 25.37, 12Z 26.01
openweather	Wind direction (wind_direction)	07-21 12Z 251.2, 18Z 259.8, 07-22 00Z 234.3, 06Z 189.3, 12Z 255.2, 18Z 179.6, 07-23 00Z 208.2, 06Z 107.8, 12Z 59.77, 18Z 106.7, 07-24 00Z 158.7, 06Z 149.2, 12Z 140.6
openweather	Wind speed (wind_speed)	07-21 12Z 2.009, 18Z 1.646, 07-22 00Z 2.314, 06Z 1.842, 12Z 1.942, 18Z 2.202, 07-23 00Z 2.64, 06Z 2.198, 12Z 3.3, 18Z 4.267, 07-24 00Z 2.437, 06Z 1.533, 12Z 1.422

Source	Metric	Values
purpleair	PM2.5 (pm25)	07-21 12Z 8.122, 18Z 7.909, 07-22 00Z 9.873, 06Z 6.863, 12Z 7.982, 18Z 11.17, 07-23 00Z 14.63, 06Z 22.42, 12Z 17.53, 18Z 9.904, 07-24 00Z 10.35, 06Z 13.53, 12Z 14.59
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	CO column (s5p_co_column)	07-21 18Z 0.02656, 07-22 18Z 0.03123, 07-23 18Z 0.02345
sentinel5p	CO granules available (s5p_co_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	07-21 18Z 0.0002243, 07-22 18Z 0.0003967, 07-23 18Z 0.0001668
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	07-21 18Z 5.873e-05, 07-22 18Z 5.036e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	07-21 18Z 3.63e-05, 07-22 18Z -2.412e-06, 07-23 18Z 5.971e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
tceq	Carbon monoxide (co)	07-21 12Z 0.16, 18Z 0.1056, 07-22 00Z 0.21, 06Z 0.1056, 12Z 0.1556, 18Z 0.1222, 07-23 00Z 0.18, 06Z 0.1333, 12Z 0.1333, 18Z 0.1611, 07-24 00Z 0.22, 06Z 0.2056, 12Z 0.3333
tceq	Nitrogen dioxide (no2)	07-21 12Z 5.148, 18Z 4.761, 07-22 00Z 10.57, 06Z 6.367, 12Z 7.011, 18Z 6.311, 07-23 00Z 7.105, 06Z 5.303, 12Z 6.175, 18Z 6.863, 07-24 00Z 7.503, 06Z 6.335, 12Z 6.817
tceq	Sulfur dioxide (so2)	07-21 12Z -0.2538, 18Z -0.1611, 07-22 00Z -0.1222, 06Z -0.2889, 12Z -0.07222, 18Z 0.07778, 07-23 00Z -0.1667, 06Z -0.3, 12Z -0.2333, 18Z -0.3278, 07-24 00Z -0.27, 06Z -0.3556, 12Z -0.3167

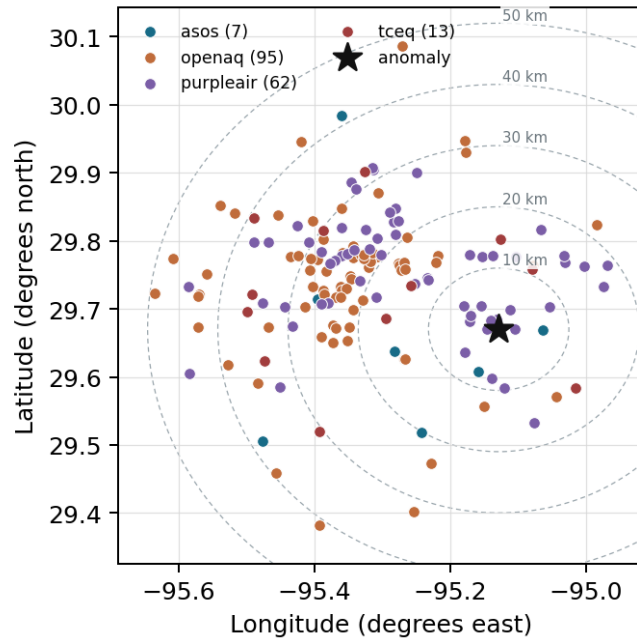
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

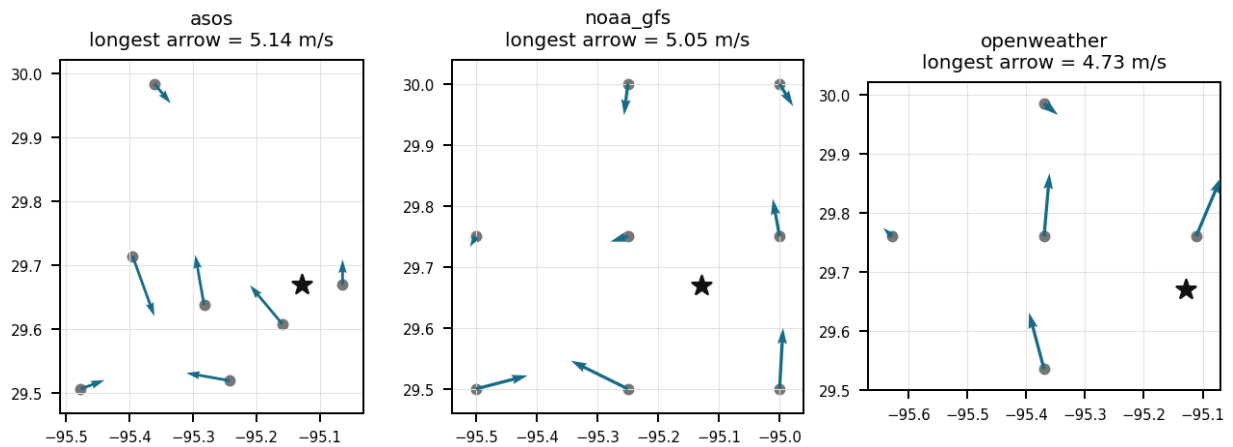
Source	Metric	Values
asos	Relative humidity (humidity)	T41 (6.224 km) 71.46, EFD (7.56 km) 71.37, HOU (15.304 km) 70.76, LVJ (20.054 km) 75.2, MCJ (26.202 km) 67.05, AXH (38.303 km) 71.77, IAH (41.517 km) 71.2
asos	Air temperature (temperature)	T41 (6.224 km) 30.18, EFD (7.56 km) 30.24, HOU (15.304 km) 30.22, LVJ (20.054 km) 29.43, MCJ (26.202 km) 30.62, AXH (38.303 km) 29.53, IAH (41.517 km) 30.22
asos	Wind direction (wind_direction)	T41 (6.224 km) 182.8, EFD (7.56 km) 175.6, HOU (15.304 km) 173.1, LVJ (20.054 km) 156.2, MCJ (26.202 km) 188.3, AXH (38.303 km) 138.7, IAH (41.517 km) 189.7
asos	Wind speed (wind_speed)	T41 (6.224 km) 3.319, EFD (7.56 km) 3.344, HOU (15.304 km) 3.565, LVJ (20.054 km) 1.951, MCJ (26.202 km) 4.166, AXH (38.303 km) 1.579, IAH (41.517 km) 3.269
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.25 (14.723 km) 5937, gfs:29.75,-95.00 (15.268 km) 5936, gfs:29.50,-95.25 (22.259 km) 5934, gfs:29.50,-95.00 (22.624 km) 5934, gfs:29.75,-95.50 (36.964 km) 5938, gfs:30.00,-95.25 (38.518 km) 5939, gfs:30.00,-95.00 (38.729 km) 5939, gfs:29.50,-95.50 (40.594 km) 5935
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.25 (14.723 km) 871.2, gfs:29.75,-95.00 (15.268 km) 601.7, gfs:29.50,-95.25 (22.259 km) 561.4, gfs:29.50,-95.00 (22.624 km) 562.8, gfs:29.75,-95.50 (36.964 km) 951.6, gfs:30.00,-95.25 (38.518 km) 804, gfs:30.00,-95.00 (38.729 km) 431.3, gfs:29.50,-95.50 (40.594 km) 549

Source	Metric	Values
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.25 (14.723 km) 50.57, gfs:29.75,-95.00 (15.268 km) 51.44, gfs:29.50,-95.25 (22.259 km) 51.74, gfs:29.50,-95.00 (22.624 km) 52.14, gfs:29.75,-95.50 (36.964 km) 49.52, gfs:30.00,-95.25 (38.518 km) 50.47, gfs:30.00,-95.00 (38.729 km) 50.99, gfs:29.50,-95.50 (40.594 km) 50.4
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.25 (14.723 km) 1009, gfs:29.75,-95.00 (15.268 km) 1010, gfs:29.50,-95.25 (22.259 km) 1009, gfs:29.50,-95.00 (22.624 km) 1010, gfs:29.75,-95.50 (36.964 km) 1008, gfs:30.00,-95.25 (38.518 km) 1008, gfs:30.00,-95.00 (38.729 km) 1008, gfs:29.50,-95.50 (40.594 km) 1009
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.25 (14.723 km) 21, gfs:29.75,-95.00 (15.268 km) 20.99, gfs:29.50,-95.25 (22.259 km) 20.81, gfs:29.50,-95.00 (22.624 km) 20.81, gfs:29.75,-95.50 (36.964 km) 21, gfs:30.00,-95.25 (38.518 km) 21.03, gfs:30.00,-95.00 (38.729 km) 21.1, gfs:29.50,-95.50 (40.594 km) 20.89
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.25 (14.723 km) 0.2075, gfs:29.75,-95.00 (15.268 km) 0.1558, gfs:29.50,-95.25 (22.259 km) 0.4516, gfs:29.50,-95.00 (22.624 km) 0.8958, gfs:29.75,-95.50 (36.964 km) 0.2608, gfs:30.00,-95.25 (38.518 km) -0.08919, gfs:30.00,-95.00 (38.729 km) 0.115, gfs:29.50,-95.50 (40.594 km) 1.218
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.25 (14.723 km) -0.4003, gfs:29.75,-95.00 (15.268 km) 0.04716, gfs:29.50,-95.25 (22.259 km) 0.008824, gfs:29.50,-95.00 (22.624 km) 0.7372, gfs:29.75,-95.50 (36.964 km) -0.4862, gfs:30.00,-95.25 (38.518 km) -0.7937, gfs:30.00,-95.00 (38.729 km) -0.8112, gfs:29.50,-95.50 (40.594 km) -0.1237
openaq	Ozone (ozone)	332 (0.0 km) 0.03439, 2800 (11.568 km) 0.02475, 2039 (14.072 km) 0.02587, 320 (14.284 km) 0.0287, 3378 (14.341 km) 0.02667, 5077791 (14.578 km) 0.02566, 3214 (14.733 km) 0.02873, 325 (16.161 km) 0.009385, +9 more
openaq	PM10 (pm10)	271 (0.0 km) 43.52, 13380082 (14.341 km) 36.62, 15852419 (21.343 km) 30.59
openaq	PM2.5 (pm25)	268 (0.0 km) 15.7, 13955815 (12.759 km) 7.832, 14404954 (13.75 km) 9.543, 14157975 (14.287 km) 10.72, 3379 (14.341 km) 11.23, 15539682 (14.345 km) 10.41, 8967123 (14.399 km) 11.45, 8967479 (14.399 km) 10.91, +67 more
openweather	Cloud cover (cloud_cover)	grid:east (10.189 km) 44.71, grid:center (25.377 km) 46.42, grid:south (27.694 km) 50.93, grid:north (42.053 km) 37.81, grid:west (49.32 km) 43.78
openweather	Relative humidity (humidity)	grid:east (10.189 km) 77.51, grid:center (25.377 km) 71, grid:south (27.694 km) 73.46, grid:north (42.053 km) 71.22, grid:west (49.32 km) 71.65
openweather	Precipitation rate (precipitation)	grid:east (10.189 km) 0.2751, grid:center (25.377 km) 0.2162, grid:south (27.694 km) 0.4288, grid:north (42.053 km) 0.2531, grid:west (49.32 km) 0.6318
openweather	Surface pressure (pressure)	grid:east (10.189 km) 1011, grid:center (25.377 km) 1011, grid:south (27.694 km) 1011, grid:north (42.053 km) 1011, grid:west (49.32 km) 1011
openweather	Air temperature (temperature)	grid:east (10.189 km) 30.15, grid:center (25.377 km) 31.31, grid:south (27.694 km) 30.4, grid:north (42.053 km) 30.83, grid:west (49.32 km) 30.89
openweather	Wind direction (wind_direction)	grid:east (10.189 km) 176.7, grid:center (25.377 km) 201.1, grid:south (27.694 km) 180.3, grid:north (42.053 km) 148.4, grid:west (49.32 km) 184.4
openweather	Wind speed (wind_speed)	grid:east (10.189 km) 3.162, grid:center (25.377 km) 2.286, grid:south (27.694 km) 2.443, grid:north (42.053 km) 2.29, grid:west (49.32 km) 1.571
purpleair	PM2.5 (pm25)	128007 (1.624 km) 6.375, 161689 (1.695 km) 12.56, 222589 (1.835 km) 12.74, 268165 (2.295 km) 13.97, 222593 (3.625 km) 12.46, 186263 (4.256 km) 13.87, 98633 (4.617 km) 2.646, 244939 (4.67 km) 10.84, +54 more
tceq	Carbon monoxide (co)	482011039 (0.032 km) 0.09846, 482011035 (14.34 km) 0.1672, 482011052 (29.74 km) 0.2061
tceq	Nitrogen dioxide (no2)	482011039 (0.032 km) 7.214, 482011015 (10.969 km) 7.925, 482011035 (14.34 km) 7.153, 482011050 (14.586 km) 4.05, 482010026 (14.755 km) 7.216, 482010416 (16.16 km) 6.67, 482011052 (29.74 km) 5.002, 480391004 (30.466 km) 5.186, +4 more
tceq	Sulfur dioxide (so2)	482011039 (0.032 km) 0.3169, 482011035 (14.34 km) -0.4778, 482010051 (33.811 km) -0.4905

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide

Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.

Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.

Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.

Missing or thin evidence means Unsure, not Invalid.

A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.

Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 35

The tceq monitor co-located with the PM10 anomaly showed low CO at 0.1 ppm, low NO2 at 8.3 ppb, and very low SO2 at 0.2 ppb at 2026-07-23T01:00:00+00:00, which argues against a strongly combustion-dominated or sulfur-rich plume as the primary cause of the PM10 spike.

Calm-wind flag: wind direction unstable under calm conditions. noaa_gfs: event wind 1.4533058882624432 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 1.1164708013951699 m/s; n=96 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 2 of 35

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 3 of 35

A sharp drop in planetary boundary layer height from 1741 meters at 18:00 UTC to 533 meters at 00:00 UTC modeled by gfs restricted atmospheric volume and trapped ground-level particulate matter.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 4 of 35

The openaq PM10 anomaly at 2026-07-23T01:00:00+00:00 occurred at 29.670, -95.129 in Houston and had a measured value of 93.0 ug/m3 versus an expected value of 49.7962962962963 ug/m3, corresponding to a z-score of 4.562483573996698 and severe classification.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 5 of 35

The anomaly occurred on July 23rd, with the highest PM2.5 levels recorded at 14:63 ug/m3.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 6 of 35

The nearest openaq PM10 sensor at the anomaly location reported 93.0 ug/m3 at 2026-07-23T01:00:00+00:00, while the 6-hour mean for openaq PM10 centered on 2026-07-23 00Z was 55 ug/m3.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 7 of 35

The elevation of PM10 relative to PM2.5 measured by openaq alongside low concurrent concentrations of NO2, CO, and SO2 reported by tceq suggests a mechanical dust source rather than an industrial combustion event.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 8 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 9 of 35

The PM2.5 levels are elevated, with a mean of 11.7877 ug/m3 and a range of 0.0-149.3 ug/m3.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 10 of 35

The PM10 anomaly at 2026-07-23T01:00:00+00:00 registered a z-score of 4.56 and was classified as severe by zscore, stl, and isolation forest algorithms.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 11 of 35

The nearest openaq PM2.5 sensor at the anomaly location reported 20.4 ug/m3 at 2026-07-23T01:00:00+00:00, and the nearest purpleair PM2.5 sensor 6.212 km away reported 11.5 ug/m3 at the same time.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 12 of 35

The temperature inversion over the Houston area is causing a buildup of pollutants near the surface.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 13 of 35

The GFS model indicated a collapse of the planetary boundary layer height to approximately 533 meters around 2026-07-23T00:00:00Z.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 14 of 35

Convective storm activity recorded by openweather and asos around 01:00 UTC on July 23 produced localized rainfall up to 11.53 mm/h and wind gusts, supporting mechanical resuspension of coarse dust prior to rain washout.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 15 of 35

The air quality anomaly in Houston, Texas is caused by high levels of SO₂ particles.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 16 of 35

A combustion-dominated or sulfur-rich industrial plume is contradicted because collocated regulatory gases remained modest near the anomaly, with CO at 0.1 ppm, NO₂ at 8.3 ppb, SO₂ at 0.2 ppb, and satellite SO₂ near zero on the prior overpass, despite somewhat elevated satellite CO and HCHO earlier that day.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 17 of 35

A nearby industrial facility or power plant is releasing excessive amounts of particulate matter and other air pollutants into the atmosphere.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 18 of 35

At 2026-07-23T01:00:00+00:00, an openaq monitoring sensor at (29.670, -95.129) recorded a PM10 concentration of 93.0 µg/m3 against an expected value of 49.8 µg/m3.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 19 of 35

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 20 of 35

The air quality anomaly in Houston, Texas on July 24th is likely caused by a combination of factors including weather patterns and local emissions.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 21 of 35

Meteorological conditions near eastern Houston around the anomaly included high humidity near 84 to 86 percent, southerly to southwesterly surface flow, low-to-moderate wind speeds, extensive cloud cover, and nearby rainfall exceeding 11 mm/h, which supports a transient shower or outflow process capable of locally lofting coarse particles.

Calm-wind flag: wind direction unstable under calm conditions. noaa_gfs: event wind 1.4533058882624432 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 1.1164708013951699 m/s; n=96 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 22 of 35

The air quality anomaly in Houston, Texas on July 24th is also characterized by elevated levels of NO₂.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 23 of 35

Rain-associated particle growth and wet mechanical splash or resuspension is a plausible contributor to the eastern Houston PM₁₀ spike because humidity near the anomaly was high at about 84 to 86 percent and openweather reported precipitation near 11.53 mm/h at nearly the same time.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 24 of 35

Rapid evening stabilization and collapse of the planetary boundary layer modeled by gfs suppressed vertical mixing, leading to localized accumulation of coarse particulate matter.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 25 of 35

An openaq monitor in Houston recorded a PM10 concentration spike of 93.0 $\mu\text{g}/\text{m}^3$ at 2026-07-23T01:00:00Z.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 26 of 35

A local coarse-particle plume from road dust, industrial material handling, construction activity, or similar mechanically generated dust is a plausible cause of the eastern Houston PM10 spike because PM10 reached 93 $\mu\text{g}/\text{m}^3$ while colocated PM2.5 was only 20.4 $\mu\text{g}/\text{m}^3$ and nearby PurpleAir PM2.5 remained modest rather than showing an equally extreme fine-particle increase.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 27 of 35

A local coarse-particle event is supported because the anomalous eastern Houston PM10 concentration reached 93 ug/m3 at 01:00 UTC while the collocated PM2.5 concentration was only 20.4 ug/m3 and nearby PurpleAir PM2.5 was 11.5 ug/m3, indicating a much stronger coarse-mode than fine-mode enhancement.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 28 of 35

Transport of coarse material from the Gulf or ship-channel sector is a plausible contributor to the eastern Houston PM10 spike because local winds near the anomaly were southerly to south-southwesterly at roughly 180 to 203 degrees with light-to-moderate speeds around 2 to 5 m/s during a humid marine air mass.

Calm-wind flag: wind direction unstable under calm conditions. noaa_gfs: event wind 1.4533058882624432 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 1.1164708013951699 m/s; n=96 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 29 of 35

Convective outflow winds associated with localized rain showers detected by openweather and asos likely re-suspended coarse surface dust, generating the PM10 spike observed by openaq.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 30 of 35

A weather pattern or storm system is bringing in air from a nearby region with poor air quality, contributing to the anomaly.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 31 of 35

OpenWeather meteorological data recorded wind speeds reaching 4.73 m/s and precipitation rates of 11.53 mm/h near the anomaly location around 2026-07-23T01:00:00Z.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 32 of 35

Fugitive dust or industrial point-source particulate emissions near the monitoring site recorded by openaq and evaluated alongside baseline combustion gases from tceq contributed to the elevated PM10 reading.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 33 of 35

Rain-related mechanical aerosol generation and resuspension are supported because humidity near the anomaly was about 84 to 86 percent, cloud cover was about 85 percent, precipitation near the site was 11.53 mm/h, and low-level winds were southerly to south-southwesterly around 180 to 203 degrees at roughly 2.1 to 4.7 m/s near the anomaly time.

Calm-wind flag: wind direction unstable under calm conditions. noaa_gfs: event wind 1.4533058882624432 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 1.1164708013951699 m/s; n=96 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 34 of 35

The openaq monitor at eastern Houston recorded a severe PM10 enhancement to 93 $\mu\text{g}/\text{m}^3$ at 2026-07-23T01:00:00+00:00 while nearby PM2.5 was only moderately elevated near 20.4 $\mu\text{g}/\text{m}^3$, which is more consistent with a coarse-particle event than with a fine-particle-dominated smoke episode.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 35 of 35

The air quality anomaly in Houston, Texas on July 24th is characterized by elevated levels of PM2.5.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.
